Sequence as a Function

A sequence can be considered as a function defined on the set of natural numbers $\mathbb{N}$. In particular, a sequence is a function $f : \mathbb{N} \to \mathbb{R}$, where $\mathbb{R}$ is the set of all real numbers.

If the sequence is of the form $a_1, a_2, a_3, \ldots$ then we can associate the function to the sequence $a_1, a_2, a_3, \ldots$ by $f(k) = a_k$, $k = 1,2,3,\ldots$

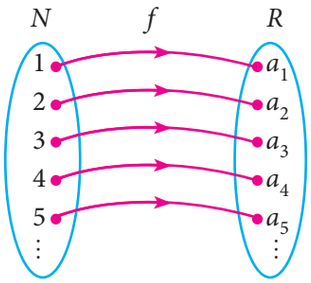


Fig 2.10

Progress Check

1. Fill in the blanks for the following sequences
   (i) 7, 13, 19, _____ , ... (ii) 2, _____, 10, 17, 26,...
   (iii) 1000, 100, 10, 1, _____, ...
2. A sequence is a function defined on the set of _____.
3. The nth term of the sequence 0,2,6,12,20,... can be expressed as _____.
4. Say True or False
   (i) All sequences are functions (ii) All functions are sequences.

Example 2.19 Find the next three terms of the sequences

(i) $\frac{1}{2}, \frac{1}{6}, \frac{1}{10}, \frac{1}{14}, \ldots$. (ii) $5, 2, -1, -4, \ldots$. (iii) $1, 0.1, 0.01, \ldots$

Solution (i) $\frac{1}{2} \underset{+4}{\rightarrow} \frac{1}{6} \underset{+4}{\rightarrow} \frac{1}{10} \underset{+4}{\rightarrow} \frac{1}{14}, \ \ldots$

In the above sequence the numerators are same and the denominator is increased by 4.

So the next three terms are $a_5 = \frac{1}{14+4} = \frac{1}{18}$

$a_6 = \frac{1}{18+4} = \frac{1}{22}$

$a_7 = \frac{1}{22+4} = \frac{1}{26}$

Note

Though all the sequences are functions, not all the functions are sequences.

(ii) $5 \underset{-3}{\rightarrow} 2 \underset{-3}{\rightarrow} -1 \underset{-3}{\rightarrow} -4, \ \ldots$

Here each term is decreased by 3. So the next three terms are $-7, -10, -13$.

(iii) $1 \underset{\sqrt{10}}{\rightarrow} 0.1 \underset{\sqrt{10}}{\rightarrow} 0.01, \ \ldots$

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